

For each problem, find a linear approximation of the given quantity.

7) $\sin 122^\circ$

8) 6.99^4

Use differentials to solve each problem.

- 9) The radius of a sphere is measured to be 7 cm, with a possible error of $\pm \frac{1}{10}$ cm. Estimate the possible propagated error in the calculated volume.

- 10) The sides of a square are measured to be 4 in, with a possible error of $\pm \frac{1}{5}$ in. Estimate the possible propagated error in the calculated area.